Code No: 126AP **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
DISTRIBUTED SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	What is Ubiquitous computing?	[2]
b)	Briefly discuss about mobile agents.	[3]
c)	Define *happened-before* relation.	[2]
d)	What is the difference between broadcast and multicast?	[3]
e)	Describe the components of datagram packet.	[2]
f)	What is marshalling and unmarshalling?	[3]
g)	What is file group and file group identifier?	[2]
h)	What is causal consistency and processor consistency?	[3]
i)	What are conflict operations? Give example.	[2]
j)	Differentiate flat and nested transactions.	[3]

PART - B

(50 Marks)

2.a) Describe briefly about the following.
i) WWW
ii) HTTP
iii) Web Services
b) Discuss significant factors affecting interacting processes in a distributed system. [5+5]

OR

3.a) How the failures are handled in distributed systems?
b) Explain about omission failures and timing failures. [5+5]

4.a) Write and explain 'snapshot' algorithm.
b) How mutual exclusion is achieved using the central server algorithm and ring based algorithm. [5+5]

OR

5.a) Write an algorithm for reliable multicasting. Explain the role hold-back queue in reliable multicasting.
b) Explain ISIS algorithm used for multicasting. [5+5]

6.a) Discuss in detail about TCP stream communication.
b) How RMI is achieved in JAVA? Explain. [5+5]

OR

7.a) Write about request-reply protocol used in client-server communication.
b) What are the characteristics of event-based systems? Explain in detail. [5+5]

8.a) Explain the operations of Andrew File System.
b) How the naming resolution is implemented in distributed systems? [5+5]

OR

9.a) What are the requirements of release consistency? How these requirements are achieved in distributed shared memory?
b) With a neat diagram, explain about write-update in distributed shared memory. [5+5]

10.a) Define shared and exclusive lock. How these locks are implemented?
b) Give brief notes on:
i) Deadlock prevention
ii) Deadlock detection. [5+5]

OR

11.a) Write and explain timestamp ordering algorithm.
b) Explain two-phase commit protocol. [5+5]

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